

FisheyeVR: Extending the Field of View by Dynamic Zooming in Virtual Reality

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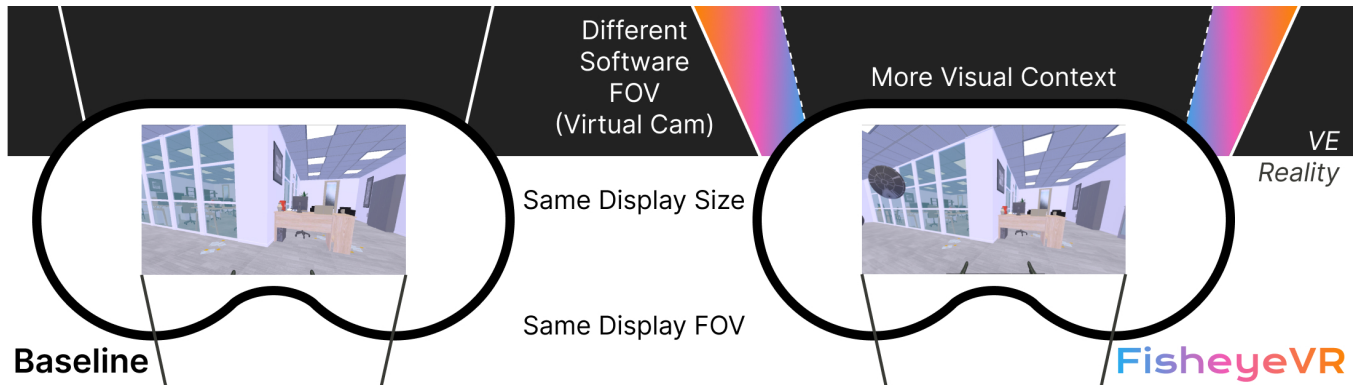


Figure 1: An illustration of FisheyeVR comparing to the baseline and the VE for study 3.

ABSTRACT

We propose *FisheyeVR*, a zooming interface in VR dynamically providing users a larger software FOV by zooming out to a shorter virtual focal length, trading in an acceptable visual distortion for more context. We conduct studies to (1) understand the visual distortion of zoom-out FOVs, (2) test 4 triggering methods with common VR scenarios and (3) evaluate the integrated *FisheyeVR* system. Our findings demonstrate that *FisheyeVR* not only significantly reduces users' physical effort and oculomotor simulator sickness but also maintains performance levels, accompanied by positive feedback.

CCS CONCEPTS

• **Hardware** → **Emerging technologies**; • **Human-centered computing** → *Interactive systems and tools*.

KEYWORDS

Virtual reality, field of view, zooming, context-aware

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1 INTRODUCTION

The field of view (FOV) is vital in virtual reality (VR) experiences. Past research has shown that large FOV in VR head-mounted displays (HMDs) offers several advantages, including reduced physical effort from head turning [3], improved user spatial ability [2, 11, 13], and enhanced sense of presence [12].

Since most commercial HMDs do not offer the same FOV of human vision [6], hardware and software approaches have been proposed to increase the FOV, e.g., adding sparse LEDs as peripheral displays [14], rendering panning [4], zooming [1] and omnidirectional content [8] in VR.

To achieve a better balance over mismatched software and display FOVs [10], we present *FisheyeVR*, a system providing a dynamic interface that adjusts the software FOV based on user command, behavior, and the VE context. *FisheyeVR*, as illustrated in Figure 1, offers a larger software FOV at opportune moments with tolerable distortion levels, thereby enhancing the overall VR experience in consumer HMDs. Figure 2 shows how *FisheyeVR* dynamically zooms with our proposed 4 triggering methods: (1) manual switching by pressing a button, (2) arms-spreading, (3) moving to bring back the normal FOV and (4) target incoming.

Our main contributions are as follows:

- We introduced a software technique for dynamically manipulating the software FOV in VR, followed by a study that offers insights into different levels of software FOVs.
- We designed and implemented a prototypical *FisheyeVR* system, then elucidated the technique's functionalities, interactions, and potential use cases through a formative study.
- We evaluated an integrated *FisheyeVR* system through a summative study, confirming its advantages in a general VR context.

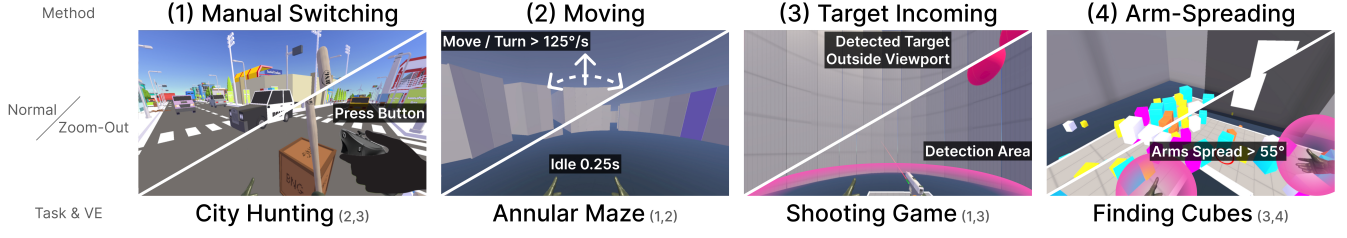


Figure 2: FisheyeVR dynamically zooms out to provide a larger software FOV. Here are illustrations of triggering methods, tasks and VEs (illustrated separately) in study 1 and 2. Number after each task name is the triggering method being tested.

2 STUDIES

We developed FisheyeVR in an iterative process, guided by 3 studies: (1) a study to understand zoom-out FOVs; (2) a formative study for the integration of triggering methods in different VR scenarios; and (3) a summative study via a 4-minute VR game, evaluating FisheyeVR’s overall impacts on users. The studies were conducted with Meta Quest 2, having display FOV of 90° [9]. **FisheyeVR only changes the software FOV using the software approach without changing the display FOV. The FOV discussed hereafter only refers to the software FOV.**

We built 4 tasks and VEs (shown in Figure 2) in Unity to represent common VR scenarios in study 1 and study 2, (1) shooting game, (2) finding cubes, (3) annular maze, and (4) city hunting. Throughout the study, we logged the direction of user’s head, positions and orientations of both hands, and the user’s position within VEs. In the study 2 and study 3, we also recorded data entries each time the zooming was activated. Meanwhile, we gathered feedback from participants via a 7-point Likert scale questionnaire, including (1) user preferences, (2) the sense of presence experienced, and assessments using (3) NASA Task Load Index (NASA-TLX) [5] and (4) Simulator Sickness Questionnaire (SSQ) [7].

2.1 Study 1: Understanding Zoom-out FOVs

To understand the tolerable distortion under large FOV, we recruited 15 participants to conduct a study. Participants were instructed to complete 4 tasks (each within 30 seconds) in random order, across 5 FOVs (90°, 110°, 130°, 150° and 170°) with HMD. The FOVs increased until the participant ask to stop; the last fully experienced FOV was their maximum acceptable FOV. Figure 3 (left) displays the

quaternion distance of head displacement under various FOVs, dots indicating participants completing tasks at each FOV. While a larger software FOV significantly reduced head’s physical effort compared to narrower ones, 130° is reported as the maximum acceptable FOV.

2.2 Study 2: Formative Study

After understanding the effects of different zoom-out FOVs, we designed 4 triggering methods to create a prototypical FisheyeVR. Given FisheyeVR’s capability to transition between the normal and zoom-out viewports, we opted for a 110° zoom-out FOV, rather than 130°, in subsequent studies to minimize the risk of excessive visual distortion that could potentially interact with the triggering methods. We recruited 13 participants for a within-subject study to get a deeper understanding. Participants were asked to perform 4 tasks (each within 1 minute) in random order with corresponded triggering methods and no FisheyeVR setting as a baseline. The center of Figure 3 shows that arm-spreading method leads to a significant reduction in head’s physical effort compared to the baseline, indicating the potential of FisheyeVR’s dynamic mechanism.

2.3 Study 3: Summative Study

Following the insights from the formative study about triggering methods under various conditions, we have integrated them into a cohesive triggering mechanism. The numbers assigned to each method in Figure 2 indicate the priority of their responses when different conditions occur simultaneously. To evaluate the efficacy of the integrated FisheyeVR, we recruited 14 participants to conduct a within-subject study. In this study, a VR game, shown in Figure 1, was developed to evaluate participants’ overall experience and reactions to FisheyeVR. This study further substantiated the benefits of FisheyeVR. Participants using FisheyeVR reported significantly reduced levels of oculomotor simulator sickness compared to the baseline, shown in the right of Figure 3.

3 CONCLUSION

Collectively, we have proposed, designed, configured, and validated a dynamic software FOV for VR HMDs that expands the available FOV and augments the user’s overall VR experience.

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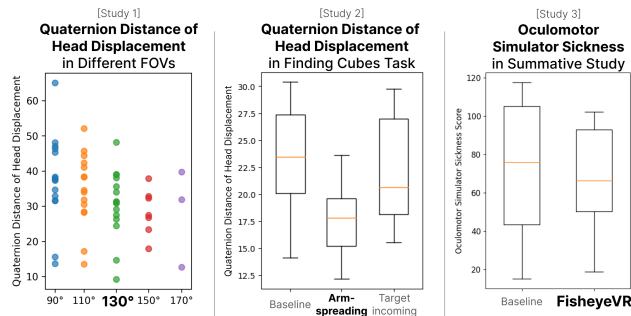


Figure 3: Results of 3 studies.

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